Lesson 12 Teacher's Key: Hearing It Happen

Godot Engine - Instructor Reference Only

VIDEO GAME CREATION

What This Lesson Teaches

No new Godot mechanism this week - dedicated AudioStreamPlayer2D nodes were already established in Lessons 6 and 8 (WinSound, LifeLostSound, GameOverSound). This lesson adds one more node in the exact same style (WaveSound) and makes GameOverSound mandatory if it was skipped - purely a matter of recognizing which existing branches are the correct trigger points.

Level 1 - Full Worked Solution

- WaveSound is the one genuinely new node this week - a dedicated AudioStreamPlayer2D living on GameManager, exactly like WinSound/LifeLostSound/GameOverSound before it.
- sfx_shoot and sfx_hit stay exactly where they were (Player's and Enemy's own AudioStreamPlayer2D nodes, from Lessons 3-4) - this lesson only touches GameManager's round-ending sounds.
- GameOverSound, optional since Lesson 8, becomes mandatory starting today - if a student skipped it, this is the lesson where it has to go in.

Why This Structure

- Keeping every round-ending sound on its own small, dedicated node (rather than one shared node swapping between clips) matches the project's established per-event pattern from Lessons 6 and 8, and keeps each node trivially simple - one node, one clip, one job.
- This is a direct continuation of Lesson 8's audio setup, not a new system - adding WaveSound is exactly the same two-step (add node, assign stream, call .play() at the right spot) students already did twice before.

Level 2 / Level 3 Talking Points

- Ask students to name the four round-ending sounds now living on GameManager (WinSound, WaveSound, LifeLostSound, GameOverSound) and, for each, the exact one-time event that triggers it.
- For students comparing engines: Unity's parallel instance.waveClip via PlayClipAtPoint() this week is the same dedicated-per-event pattern, just reached through a static singleton instead of direct node references.

Common Student Mistakes

- Placing the $WaveSound.play() call outside the else branch, causing it to also fire on the true win.
- Node-not-found errors from a typo'd or misplaced WaveSound node name/path - it must be a direct child of GameManager, named exactly WaveSound.
- Skipping GameOverSound because it was optional before - it's mandatory as of this lesson.